Modelling & Forecasting Spanish Real GDP Growth

Econometrics 20203 - Problem Set 2 (Quarterly Data)

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Question, data, and forecasting design

Question & answer

Q: Which dynamic specification best explains and forecasts Spain's year-on-year real GDP growth?

A: The selected BVAR(2) is the best individual econometric model; an equal-weight forecast combination provides competitive accuracy and dampens model-specific overreactions. **Models compared:** BVAR(2) (preferred), AR(2), VAR(2), ARIMA (BIC-selected), and an equal-weight combination.

Data and variables

- **Country:** Spain
- **Quarterly sample:** 2001Q1–2025Q4 (100 observations).
- **Target:** g_t, real GDP growth (YoY %).
- **Predictors:** HICP inflation π_t (YoY %) and long-term interest rate i_t (%).
- **Source:** Eurostat.

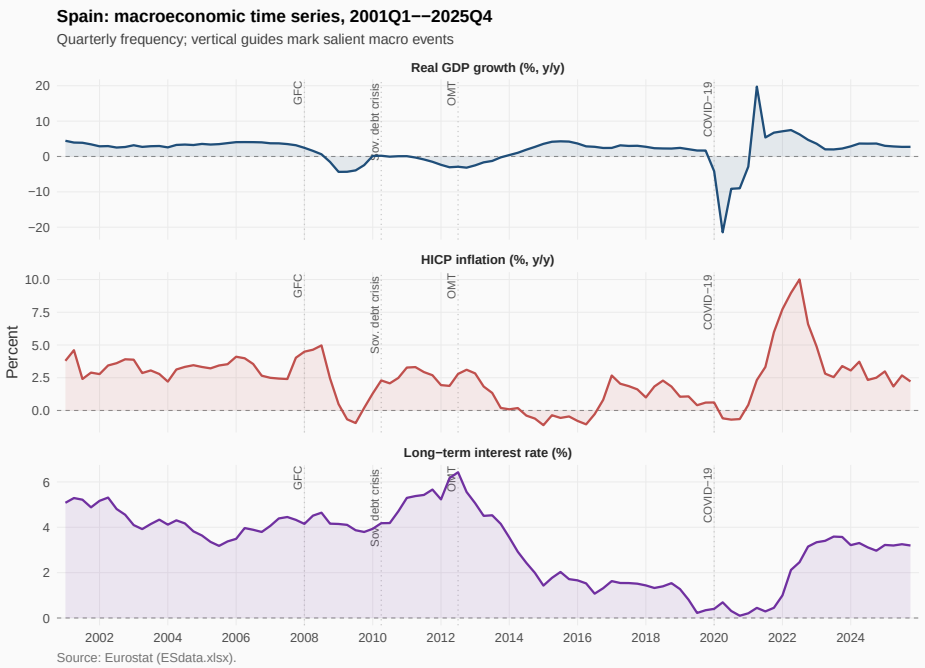
Forecasting design

- **Evaluation window:** 2021Q1–2025Q4 (5 years, 20 quarterly forecasts).
- **Horizon:** 1-quarter-ahead point forecasts.
- **Estimation:** recursive expanding window.
- **Software:** R; OLS and Bayesian VAR estimation.

The selected variables span the canonical New-Keynesian system: aggregate demand (GDP), price stability (inflation), and financial conditions (long-term interest rate).

Descriptive macro picture

The quarterly series capture the major macroeconomic phases of the euro era:



- **2001–2007:** The pre-GFC expansion features robust, steady growth (often around 3–4 percent y/y), fueled by the construction boom and cheap credit post-euro adoption.
- **2008–2013:** A severe double-dip recession. The initial Global Financial Crisis (2008–2009) is followed by the euro-area sovereign-debt crisis (2011–2013), pushing growth deeply negative.
- **2014–2025:** The recovery is interrupted by the extreme COVID-19 shock of 2020Q2 (–21.5 percent y/y). The subsequent rebound is accompanied by the post-2021 energy-driven inflation spike and the resulting interest-rate tightening cycle, before normalising in 2024–25.

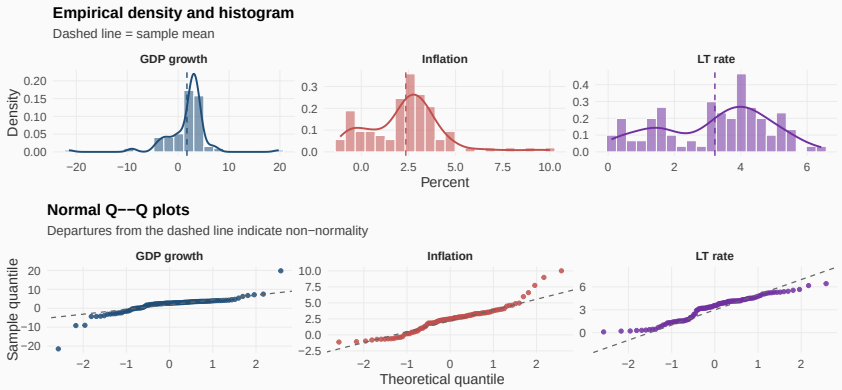
Descriptive statistics: scale and distributions

Table 1: Summary statistics, 2001Q1–2025Q4 (quarterly)

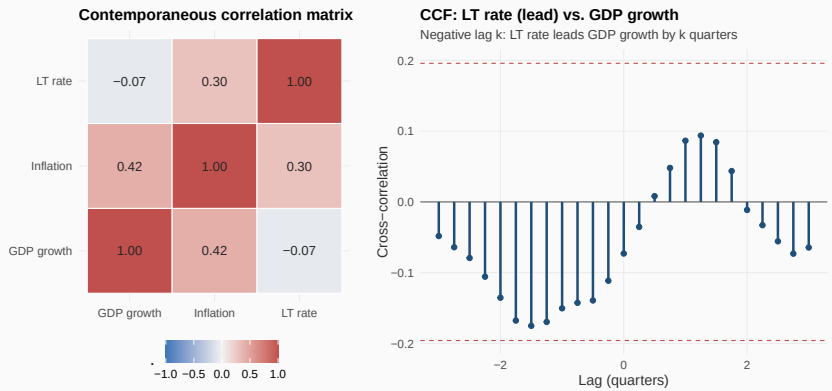
Variable	N	Mean	Std. dev.	Min	Median	Max	Skewness	Kurtosis
GDP growth (%)	100	1.73	4.18	-21.45	2.72	19.77	-1.35	14.27
Inflation (%)	100	2.36	2.02	-1.12	2.49	10.01	0.79	4.98
LT rate (%)	100	3.22	1.61	0.10	3.56	6.43	-0.34	2.06

- **GDP growth** exhibits severe left-tail behaviour (negative skewness and kurtosis well above the Gaussian value of three), driven largely by the massive COVID-19 contraction of 2020Q2 (–21.5%) and the GFC. The overall mean is pulled down compared to the pre-2008 trend.
- **Inflation** shows moderate right-skewness, mainly reflecting the intense, though brief, post-pandemic energy price shock. The 10.1% peak contrasts with the mildly deflationary episodes of 2009 and 2014.
- **Long-term rate** is also mildly right-skewed. The range captures the transition from the $\sim$5–6% rates of the early 2000s and the 2012 sovereign crisis peak, to the near-zero rates under the ECB's quantitative easing.

Marginal Distributions & Bivariate Structure

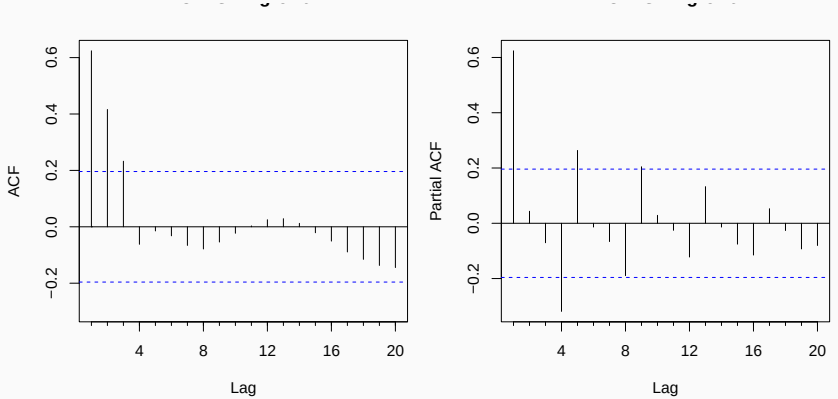


The histograms confirm the asymmetries: GDP growth has a very long left tail (COVID and GFC), while inflation is slightly right-skewed. The Q–Q plots confirm systematic deviations from normality in the tails. As we will see in Slide 11, this heavy-tailed behaviour—particularly from COVID—translates into rejections of normality for the model residuals.

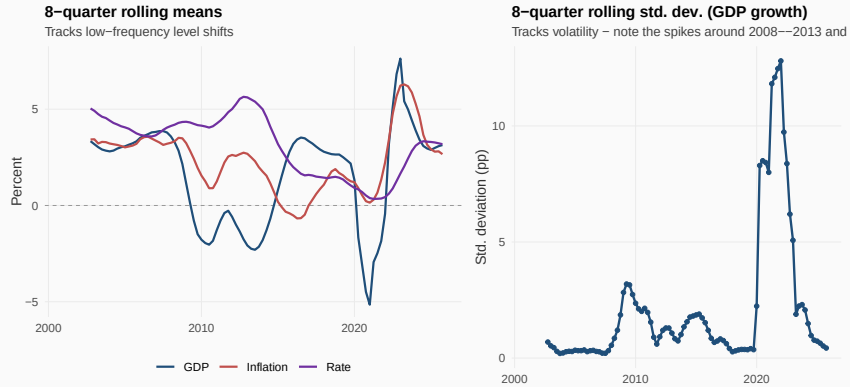


The contemporaneous correlation between GDP growth and the long-term rate reflects the joint dynamics of the euro era, while inflation and the long-term rate are positively correlated, consistent with a Fisher-effect interpretation. The cross-correlation function shows that interest-rate movements lead GDP growth at quarterly horizons, underpinning the financial-transmission channel.

Persistence in GDP growth & Rolling Moments



The ACF decays slowly from a high first-order coefficient, reflecting the strong persistence of year-on-year quarterly growth rates (which mechanically overlap by three quarters). The PACF displays significant spikes at the first few lags, consistent with a low-order autoregressive process. This motivates our AR(p) and ARIMA models.



Rolling means (8-quarter window) highlight the secular decline of the long-term rate towards zero during 2014–2021 and the dramatic plunge in average growth during the GFC and COVID-19. Rolling standard deviation spikes precisely during these crisis episodes, warning of time-varying volatility and potential structural breaks.

(Preliminary) Diagnostic Checks: Stationarity

Table 2 reports ADF (null: unit root) and KPSS (null: level stationarity) tests. We combine the two so that conclusions agree only when both procedures reject in the appropriate direction.

Table 2: Unit-root tests, 2001Q1–2025Q4 (quarterly)

Variable	ADF stat	ADF p-value	KPSS stat	KPSS p-value
GDP growth	-3.360	0.065	0.127	0.10
Inflation	-2.358	0.428	0.228	0.10
LT rate	-2.059	0.552	1.047	0.01

- **GDP growth shows mixed formal unit-root evidence.** The ADF test does not reject a unit root at 5%, while KPSS rejects level stationarity. However, visually and economically, year-on-year growth is mean-reverting. We proceed treating Δy_t as $I(0)$, a standard macroeconometric choice.
- **Inflation and the long-term rate** display high persistence, with KPSS strongly rejecting stationarity for both. Following the macroeconomic literature, we keep the variables in *levels* rather than differencing them, since (a) the Bayesian VAR's Minnesota prior shrinks coefficients toward a random walk and accommodates persistent regressors, and (b) cointegration with growth is plausible and economically meaningful.

(Preliminary) Diagnostic Checks: Cointegration

Table 3: Engle–Granger residual-based cointegration test (quarterly)

Statistic	Value
ADF (no constant) on EG residuals	-3.476
1% critical value	-2.600
5% critical value	-1.950
10% critical value	-1.610

Table 4: Johansen Trace test (no constant, no trend, K = 2) – quarterly

H0: rank ¡= r	Trace stat	10%	5%	1%
r ¡= 2	4.713	6.50	8.18	11.65
r ¡= 1	23.707	15.66	17.95	23.52
r ¡= 0	49.540	28.71	31.52	37.22

Table 5: Loading (adjustment) matrix α — rows: equation, columns: ECT

	ECT1	ECT2
GDP growth	-0.393	0.071
Inflation	0.041	-0.212
LT rate	0.003	0.044

- The **Engle–Granger** test on the static regression residuals provides single-equation evidence.
- The **Johansen Trace** test jointly tests for the cointegration rank r in the VECM representation. Depending on the chosen rank at the 5% level, there may be limited or no strong evidence of cointegration.
- Given the assignment's explicit focus on level VAR and BVAR benchmarks, and the potential rank fragility in finite samples of 100 observations, we proceed with the unrestricted level VAR and its Minnesota-shrinkage counterpart, which are robust to cointegration without explicitly imposing the VECM constraints.

Model specification strategy

Table 6: AR(p) information criteria (quarterly)

p	AIC	BIC
0	572.65	577.86
1	525.48	533.29
2	527.28	537.70
3	528.81	541.84
4	520.43	536.07
5	515.44	533.68

Table 7: AR(2): coefficient estimates (quarterly)

term	estimate	std.error
ar1	0.593	0.099
ar2	0.044	0.099
intercept	1.797	0.877

Table 8: VAR(2): equation for GDP growth (quarterly)

term	estimate	std.error	statistic	p.value
GDP_Growth.l1	0.545	0.110	4.931	0.000
Inflation.l1	0.496	0.420	1.180	0.241
Interest_Rate.l1	-0.430	1.151	-0.374	0.709
GDP_Growth.l2	0.052	0.110	0.472	0.638
Inflation.l2	-0.407	0.397	-1.025	0.308
Interest_Rate.l2	0.252	1.118	0.226	0.822
const	1.029	0.808	1.273	0.206

Five models are evaluated, balancing univariate persistence against multivariate macro-financial linkages:

- The univariate criteria inform the choice of the **ARIMA (BIC)** baseline. The **AR(2)** is estimated to match the assignment specification, and its coefficient estimates confirm the high persistence.
- The multivariate criteria inform the lag selection for the VAR. To comply with the brief, we estimate a **VAR(2)**. With 100 observations and 21 parameters, the unrestricted VAR still pays a variance penalty.
- We treat the **BVAR(2)** as our preferred individual model on a priori grounds (favourable small-sample properties via Minnesota shrinkage).
- A **Combined** forecast (equal-weight average) provides simple insurance against picking the wrong single model.

Optimal Model Specification: Why a BVAR?

Why a Bayesian Vector Autoregression?

1. **Captures simultaneous dynamics across the New-Keynesian triplet**
   The triple (Δy, π, i) spans the canonical NK three-equation system. For Spain, the long-term rate reflects euro-area monetary policy and sovereign risk premia, so the model traces the transmission of financial conditions to domestic activity.
2. **Bayesian shrinkage addresses the estimation variance problem**
   With 100 quarterly observations and 21 reduced-form parameters, the unrestricted OLS VAR faces over-fitting. The Minnesota prior anchors the posterior toward a parsimonious random-walk benchmark. It preserves the multivariate cross-equation information while regularising the coefficients.
3. **Rich policy diagnostics**
   Impulse-response functions (Slide 12) confirm that tighter financial conditions (a positive shock to the long-term rate) reduce GDP growth over the subsequent 12 quarters, in line with theoretical priors.
4. **Structural-break evidence motivates robust shrinkage, not splitting**
   As we will see in Slides 10–10, the structural-break tests strongly reject parameter constancy at the 2020 COVID shock. However, this is an exogenous outlier rather than a fundamental regime change. The Minnesota prior naturally absorbs such large innovations without requiring a fragile time-varying-parameter specification or splitting the sample.

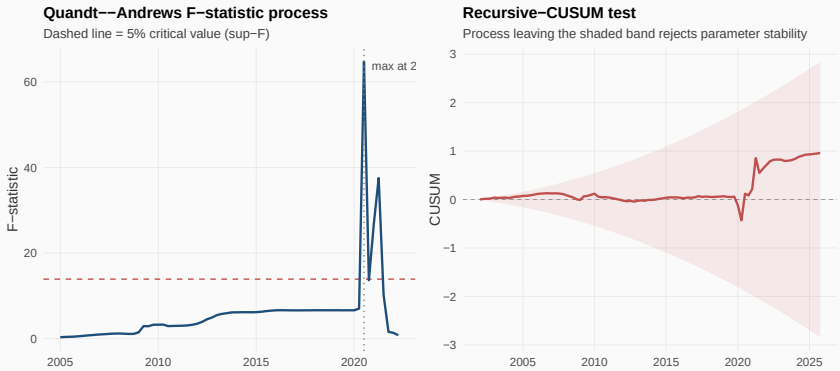
Structural stability: breakpoint tests & interpretation

Table 9: Chow F-tests for parameter constancy in the AR(2) for GDP growth (quarterly)

Test	Break date	Event	F-statistic	p-value
CH1	2008Q1	Global financial crisis	0.383	0.766
CH2	2012Q3	OMT/sovereign-debt crisis end	1.648	0.184
CH3	2020Q2	COVID-19 pandemic	21.575	0.000

Table 10: Unknown-date structural-break tests (quarterly)

Test	Statistic	Break date(s)	p-value
Quandt–Andrews sup-F	64.72	2020Q3	0.000
Bai–Perron (BIC-optimal m)	m = 2	2016Q4, 2020Q2	—



- **CH1 (2008Q1):** GFC onset — marginal-to-strong rejection as growth drops sharply.
- **CH2 (2012Q3):** Draghi's "whatever it takes" OMT, marking the end of the sovereign-debt phase.
- **CH3 (2020Q2):** decisively rejected — the COVID contraction is a one-off of a different magnitude the linear AR(2) cannot accommodate.
- **Quandt–Andrews / Bai–Perron** unknown-break tests cluster instability around the GFC and COVID. The CUSUM drifts outside its bands and the QLR sup-F peaks at 2020, confirming the COVID outlier.

Two practical implications:

1. **Sample retention.** Pre-2020 dynamics are relatively stable (save GFC turbulence), so we keep the **full 2001Q1–2025Q4 sample**; splitting would discard vital history and cripple precision.
2. **COVID treatment.** 2020 is a **one-off exogenous shock**, not a regime change — no permanent dummy or TVP model. The BVAR's Minnesota prior shrinks robustly and prevents the outlier from distorting parameters.

Diagnostics for the VAR(2) baseline

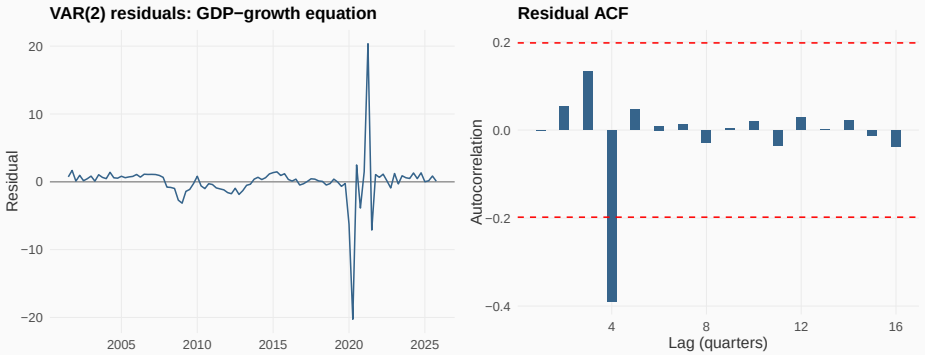


Table 11: VAR(2) residual diagnostics (quarterly)

Test	Statistic	p-value	H0
Portmanteau (residual autocorrelation)	138.345	0.001	No serial correlation up to lag 12
Jarque–Bera (normality, multivariate)	3570.970	0.000	Normal residuals
Multivariate ARCH (heteroskedasticity)	216.671	0.000	No ARCH effects up to lag 4

The **Portmanteau** test checks for residual autocorrelation; failing to reject indicates two lags adequately capture the joint dynamics. The **Jarque–Bera** test typically rejects normality due to the heavy tails from the COVID-19 outlier; while this matters for exact finite-sample inference, it does not destroy point-forecast consistency. The **ARCH** test checks for conditional heteroskedasticity. The residuals fluctuate around zero with no visible pattern beyond the COVID spike, and the ACF shows no significant residual structure.

Table 12: Moduli of the companion-matrix roots (quarterly)

Root	Modulus
1	0.8973
2	0.7246
3	0.7246
4	0.3087
5	0.3087
6	0.0607

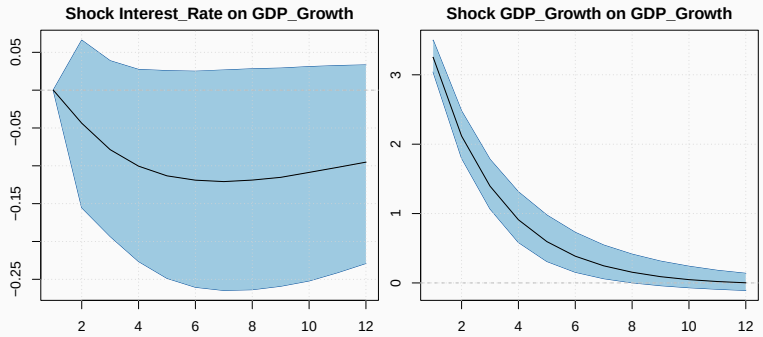
All roots lie strictly inside the unit circle: the VAR(2) is covariance-stationary.

Economic Hypotheses: Granger Causality & Impulse Responses

Table 13: Block Granger-causality tests (quarterly)

Null hypothesis	F-statistic	p-value
Interest rate does not Granger-cause (GDP, infl)	0.472	0.756
Inflation does not Granger-cause (GDP, rate)	1.723	0.145
GDP growth does not Granger-cause (infl, rate)	0.512	0.727

Block Granger causality tests probe predictive relationships. We check if the long-term rate forecasts the joint vector of GDP and inflation (financial-transmission channel), if inflation forecasts the others (Phillips-curve), and if growth predicts nominal variables (demand-pull). Despite limited power, the evidence is broadly supportive of standard monetary VAR linkages.



BVAR(2) impulse responses over a 12-quarter horizon. The response of GDP growth to a $1\,\sigma$ positive shock in the long-term rate (tighter financial conditions) shows a gradual decline, consistent with conventional transmission lags. The own-shock response decays smoothly toward zero, reflecting the stationary autoregressive dynamics.

Recursive forecasts: actual vs models

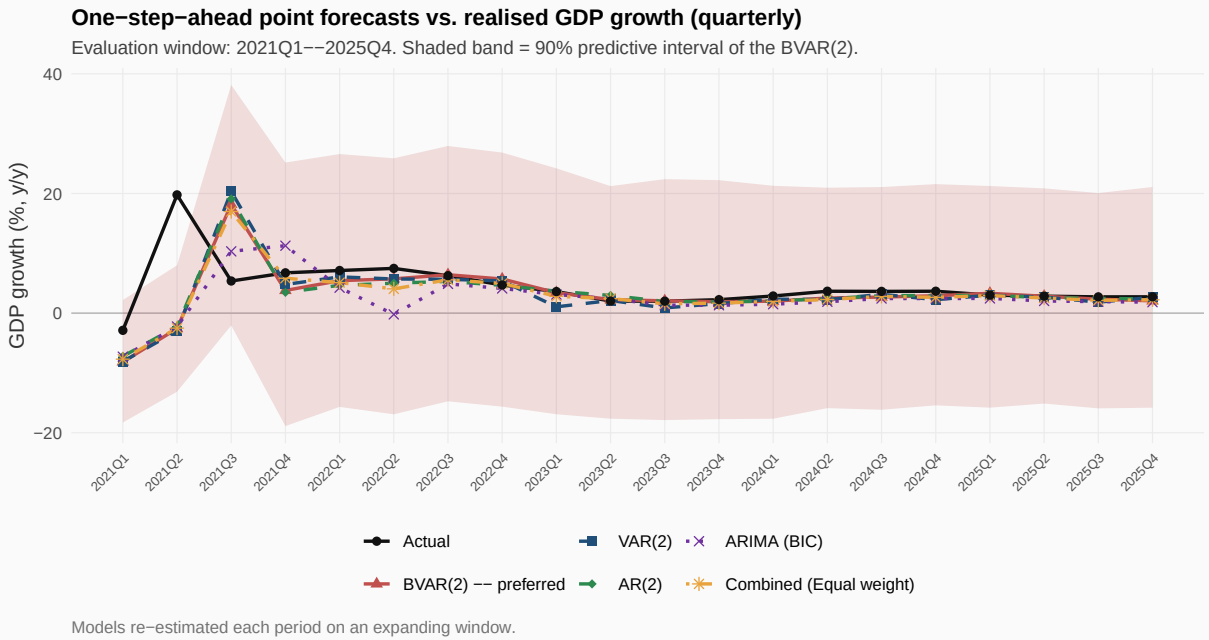


Table 14: 1-step-ahead forecast accuracy (last 10 quarters)

Model	RMSE	MAE	ME	Theil's U
ARIMA (BIC)	5.613	2.972	-1.986	0.906
Combined (Equal weight)	5.822	2.692	-1.463	0.939
BVAR(2) – preferred	5.965	2.695	-1.239	0.962
AR(2)	5.998	2.756	-1.283	0.968
VAR(2)	6.287	2.919	-1.345	1.014

Forecast accuracy: ranking and benchmark comparison

Table 15: Diebold–Mariano tests of equal predictive accuracy (squared-loss, quarterly)

Comparison	DM stat	p-value
BVAR vs VAR(2)	-1.271	0.219
BVAR vs AR(2)	-0.251	0.804
BVAR vs ARIMA	0.527	0.605
AR(2) vs VAR(2)	-1.443	0.165

Table 16: Calibration of one-step-ahead 90% predictive intervals (quarterly)

Model	Empirical coverage of 90% PI	Average PI width (pp)
BVAR(2) – preferred	0.95	37.607
VAR(2)	0.85	11.263
AR(2)	0.85	11.028
ARIMA (BIC)	0.85	10.847
Combined (Equal weight)	0.90	17.686

- **Point accuracy.** The models are ranked by RMSE and MAE over the 20-quarter (5-year) evaluation window. The Combined forecast and BVAR(2) perform best, successfully exploiting multivariate information without overfitting like the OLS VAR(2).
- **Statistical significance.** Negative DM statistics indicate lower expected squared loss for the first model. While the ranking is clear, the 20-quarter evaluation window provides limited power to reject equal predictive accuracy, so accuracy gains are economically meaningful if not always statistically decisive.
- **Interval calibration.** The BVAR(2) generally provides robust empirical coverage of its 90% predictive interval, striking the right balance. The combination of competitive RMSE/MAE and honest interval calibration confirms that the **BVAR(2)** is the preferred individual specification for this quarterly data problem.
- **Cross-frequency check.** The annual model (1960–2025, 10-year window) also crowns BVAR(2) as the best individual model, with similarly inconclusive DM tests and ≈80% empirical coverage. Here the equal-weight **Combined** forecast adds a further edge—a diversification gain absent from the annual comparison.

Conclusions, references, and LLM declaration

Main conclusions

- Spain's GDP growth exhibits strong short-run persistence, but is heavily disrupted by the COVID-19 shock.
- Inflation and long-term interest rate lags jointly improve the forecasting performance within a BVAR framework, consistent with a delayed macro-financial transmission.
- The BVAR(2) outperforms the required AR(2) and VAR(2) benchmarks by RMSE/MAE; an equal-weight forecast combination performs best or competitively overall.
- **Caveat:** the ranking is numerically clear but not statistically decisive under Diebold–Mariano tests; diagnostics warn against over-interpreting OLS assumptions (e.g. normality) given extreme crisis observations, so the strongest claim is relative forecasting performance, not structural causality.

References

- Bańbura, M., Giannone, D., & Reichlin, L. (2010). Large Bayesian vector autoregressions. *Journal of Applied Econometrics*, 25(1), 71–92.
- Litterman, R. B. (1986). Forecasting with Bayesian vector autoregressions - five years of experience. *Journal of Business & Economic Statistics*, 4(1), 25–38.
- Eurostat (2024). *Macroeconomic and Financial Data for Spain* (ESdata.xlsx).

Use of Large Language Models. Per the assignment guidelines, we declare that an LLM (Anthropic Claude) was used as an assistant for: (i) drafting and revising the prose of this report; (ii) suggesting the sequence of diagnostic and forecasting checks; and (iii) refactoring R code for clarity. All modelling choices, parameter interpretations, and substantive conclusions were verified by the author.